

PMO-CTSB — System Workflow Guide

Cybersolution Technologies · Helpdesk → Backlog → Tasks · June 2026

This document explains **how the whole system fits together**: who does what, how work moves from a client ticket to delivery and billing, and where each module lives in the app.

Audience: PMO officers, team leads, developers, support staff, and stakeholders learning the platform.

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1. What the system is for

PMO-CTSB is a **project portfolio and delivery platform** designed for government and PBT-style engagements. It connects:

- **Client agencies** (DBKL, MOH, universities, etc.)
- **Delivery projects** (contracts, letters of offer, purchase orders, tenders)
- **Operational support** (helpdesk / eTicket)
- **Planned work** (product backlog and tasks)

- **Team schedule** (calendar, site visits, UAT, go-live)
- **Billing** (phase milestones and finance views)

The intended mental model:

Intake on Helpdesk → Prioritize on Backlog → Execute as Tasks → Track on Timeline → Schedule on Calendar → Bill via Delivery/Finance

2. Users and roles

Role	Typical person	What they do in the system
Admin	IT / system owner	Users, settings, audit history, full access
PMO	Project manager, delivery lead	Create projects, assign helpdesk, manage backlog, promote to tasks, calendar, reports
Finance	Accounts / billing	Finance module, payment milestone visibility
Team member (user)	Developer, BA, QA, support	My Work , assigned tickets/backlog/tasks, view calendar
HR	(optional)	User support

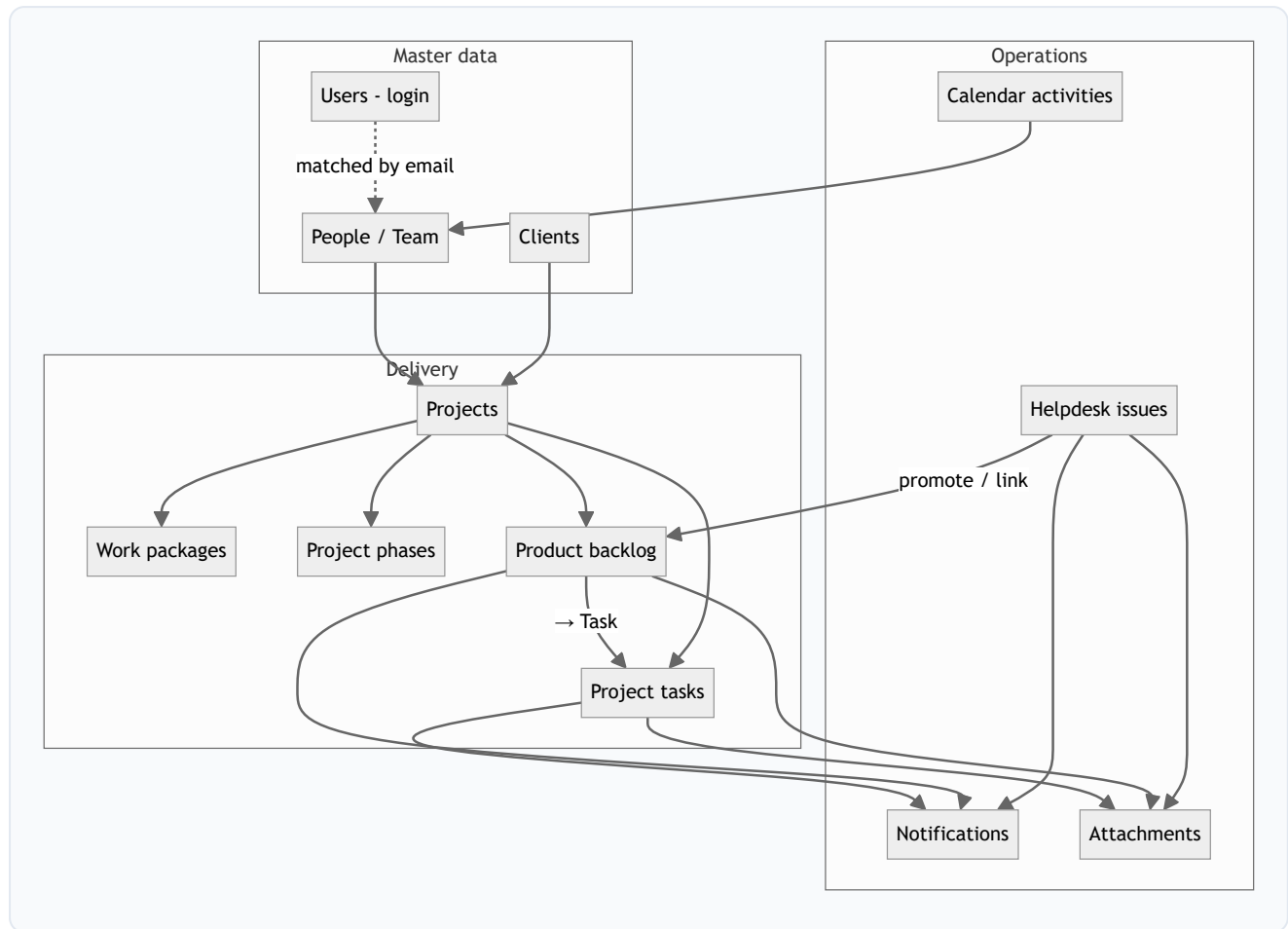
Team roster vs login accounts

- **People** (**Team** page) = delivery roster (name, email, role).
- **Users** (**System users**) = login accounts.
- The system matches them by **email** or **name** so notifications and “My Work” know who you are.

Project roster

People are assigned to projects via **project assignments** (role on project, allocation %). This roster is used for backlog @mentions and project visibility.

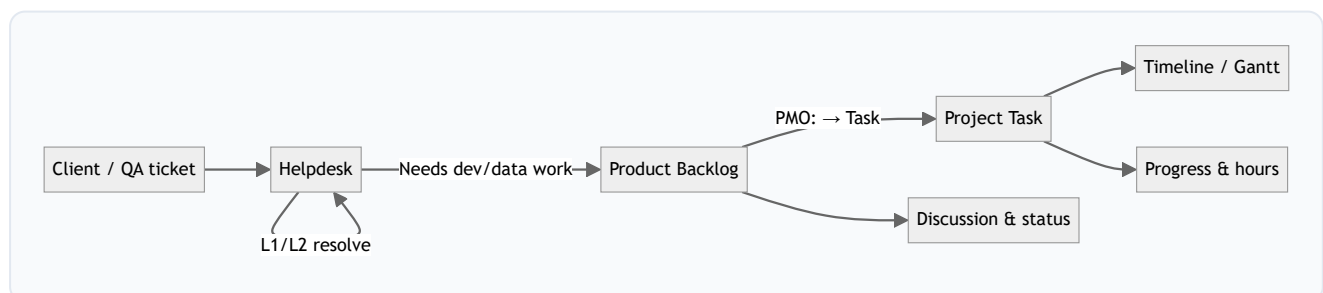
3. Core data relationships



One project can have many work packages, backlog items, tasks, phases, and linked helpdesk tickets.

4. The main delivery pipeline

This is the **most important flow** to understand.



Stage	Where	Purpose
Intake	Helpdesk	Log, assign, escalate, resolve at support level
Planning	Product backlog	Prioritize, estimate hours, assign owner
Execution	Tasks	Plan dates, track progress, complete work
Visibility	Gantt / Timeline	Portfolio and project schedules
Operations	Calendar	Meetings, UAT, FAT, outstation, go-live

Rule of thumb:

- If support can **fix it on the ticket** → stay on Helpdesk (resolve).
- If it needs **development, data work, or formal delivery** → **Backlog** → **Task**.

5. Helpdesk (eTicket)

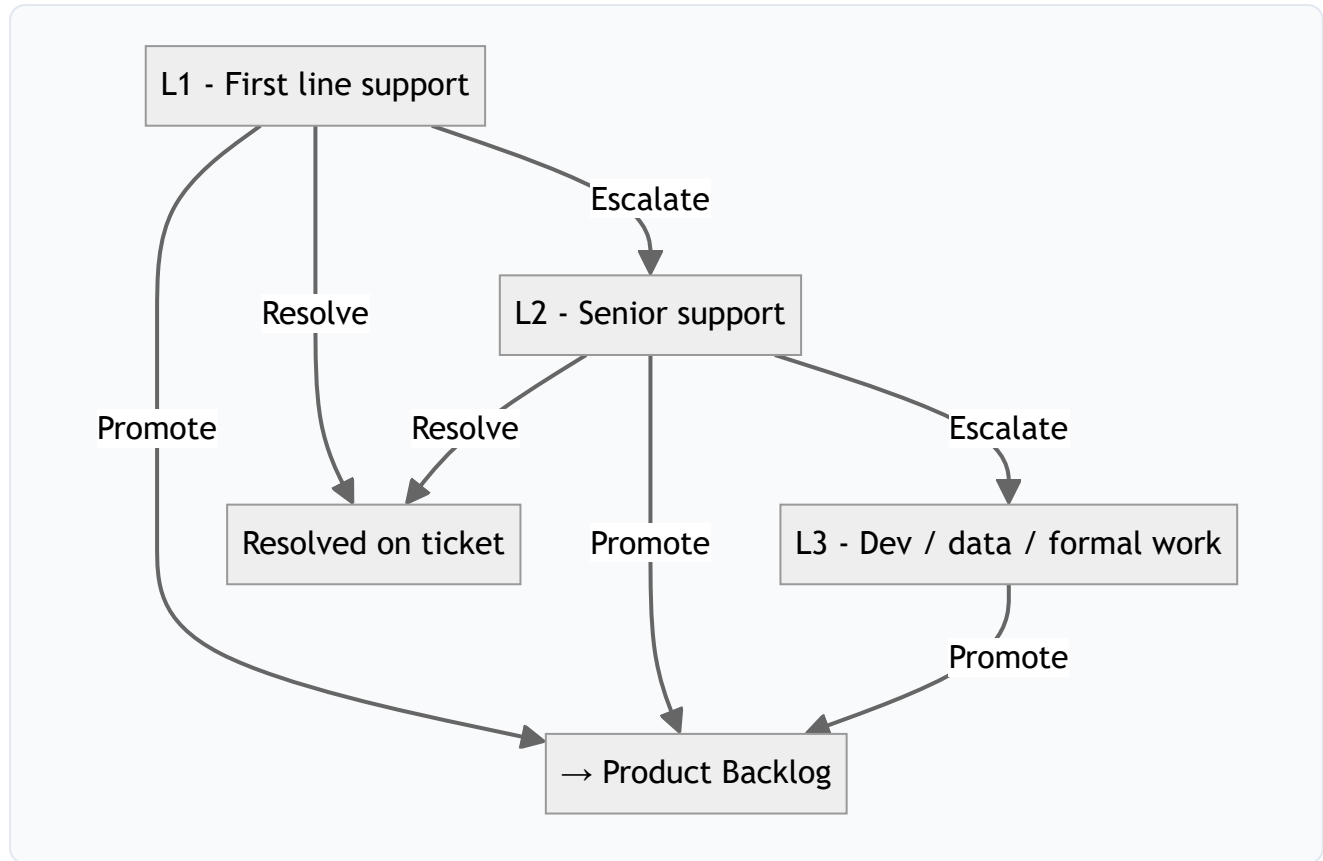
URL: /helpdesk

Purpose: Operational support queue aligned with legacy **eTicket** fields (ticket number, module, client, external ref, support level, etc.).

Ticket statuses

Status	Meaning
Open	New or not yet worked
In progress	Assignee is working on it
Waiting agency	Blocked on client / agency
Resolved	Closed at the current support level

Support levels (L1 → L2 → L3)



Level	Typical handling
L1	Password resets, how-to, basic troubleshooting
L2	Deeper analysis, configuration, recurring issues
L3	Defects, data fixes, changes needing backlog/task tracking

Who can promote Helpdesk → Backlog?

- PMO / Admin
- Ticket **reporter** (person who logged it)
- Any **rostered team member**

L1 can promote **directly** to backlog without going through L2/L3 first (when work clearly needs dev/data).

Promote to backlog — what happens?

1. User selects target **project** (if not already linked).
2. System **creates** a new backlog item **or links** to an existing one (see [Linking rules](#)).
3. Issue status moves to **in progress** (linked to delivery).
4. **Attachments** copy from issue → backlog.
5. **Assignee** on the backlog is notified.
6. **Creator** (who promoted) is stored for later status notifications.

Helpdesk notifications

Event	Who is notified
Ticket assigned	Assignee
Promoted to backlog	Backlog assignee

6. Product backlog

URL: `/projects/{id}?tab=backlog`

Purpose: Prioritized work list per project — original scope, CRs, bugs, enhancements, data work, support items.

Backlog item types

Type	Use for
Original scope	Contract / URS work
Change request (CR)	Approved CR
Bug / Defect	Functional or quality issues
Enhancement	Improvements
Data	Migration, cleansing
Support	Technical inquiries carried into delivery
Recurring	Repeat issues

Backlog statuses

Status	Meaning	Typical next step
Open	Queued, not started	Assignee picks up → In progress
In progress	Work underway	Complete → Fixed
Fixed	Work done, awaiting verification	PMO/client confirms → Closed
Closed	Verified or no further action	Archive

Permissions

Action	PMO	Assignee
Create backlog item	✓	—
Set assignee, priority, estimates	✓	—
Update status	✓	✓
Update actual hours	✓	✓
Post comments, @mention team	✓	✓ (if on project roster)
→ Task (promote to task)	✓	—
Attach files / links	✓	✓ (view); upload per module rules

Discussion and @mentions

Each backlog item has a **comment thread**. Type @ + team member name to mention someone on the project roster.

Event	Who is notified
Backlog created / assigned	Assignee
Status changed	Creator (who created or promoted the item)
New comment	Creator, assignee
@mention in comment	Mentioned person

Deep link: `/projects/{projectId}?tab=backlog&backlog={id}`

Promote backlog → Task

PMO clicks → **Task** when work is ready for scheduling and tracking.

1. Creates a **project task** (title, hours, assignee, `backlog_id`).
2. Sets backlog to **In progress** and stores `task_id` .
3. Copies **attachments** backlog → task.
4. Notifies **task assignee** (in-app; email if SMTP configured).

7. Project workspace

URL: `/projects/{id}`

Each project is a **delivery workspace** with tabs:

Tab	Purpose
Overview	Health, KPIs, charts, summary
Work packages	Split large projects (Portal, API, Migration, ...)
Backlog	Product backlog for this project
Tasks	Task list and hierarchy
Delivery	Phases (URS, UAT, FAT, go-live) and payment status
Timeline	Mini Gantt for this project
People	Who is assigned and allocation %

Work packages

Use when one contract has **multiple delivery streams**. Filter backlog and tasks by package.

Engagement types

Projects can be tagged as Contract, Letter of offer, Purchase order, Tender, etc. — affects how phases and finance are interpreted.

8. Tasks and timeline

URLs:

- Project tasks: `/projects/{id}?tab=tasks`
- Portfolio Gantt: `/gantt`
- Project timeline: `/projects/{id}?tab=timeline`

Task statuses

Status	Meaning
New	Not started
Ongoing	In progress
Done	Complete (typically 100% progress)

Tasks can be:

- **Leaf tasks** — real work items with assignee, dates, hours.
- **Group tasks** — containers for subtasks (rollup progress).

Relationship to backlog

- Task may have `backlog_id` — created from → **Task** on backlog.

- Completing the task does **not** auto-close backlog; PMO/assignee updates backlog status separately (Fixed → Closed).

Gantt and timeline

- **Gantt** — cross-project view of planned vs actual bars.
 - **Timeline** — project-scoped schedule with phases and tasks.
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9. Calendar and activities

URL: `/calendar`

Purpose: Team schedule — not the same as project tasks, but complements them.

Activity types

Meeting, Outstation, UAT, URS, FAT, DEMO, Training, Go-live, Tender, Other.

Who can edit?

PMO and Admin can log activities, import Excel, email team schedule.

Features

- Month view with type filters and KPIs (activity count, active days).
- Click a **day** to pre-fill a new activity (PMO).
- Click an **event** for details, edit, delete, resend email.
- Multi-assignee activities show as **one chip** on the calendar (grouped).
- Import historical schedules from **Excel/CSV**.

Link to My Work

My Work → **Today** shows calendar activities for the logged-in user.

10. My Work — personal queue

URL: `/my-work`

Purpose: Single place for each team member to see **their** work.

Tab	Shows
All	Combined queue
Tasks	Assigned project tasks
Backlog	Assigned open backlog items
Helpdesk	Assigned open tickets
Today	Today's calendar activities

Items are sorted by urgency (overdue tasks, priority, dates). Each row links to the right project or helpdesk detail.

11. Delivery, finance, and payments

Delivery tab (`/projects/{id}?tab=delivery`)

Tracks **project phases** — e.g. URS, Development, SIT, UAT, FAT, Go-live, Warranty.

Each phase can have:

- Target / completed dates
- Progress %
- **Payment milestone** — amount, invoice no, paid date, payment status

Finance module (`/finance`)

Roles: PMO, Admin, Finance.

Portfolio view of:

- Ready to bill
- Invoiced
- Paid & claimed
- Maintenance renewal follow-up

Finance reads from **phase payment fields** on projects — it does not replace your accounting system; it tracks milestone status for PMO.

12. Notifications

Bell icon (top bar) — in-app notifications.

Type	Trigger
Issue assigned	Helpdesk assignee set
Backlog assigned	New backlog or assignee changed
Backlog status	Creator notified when assignee changes status
Backlog comment / @mention	Creator, assignee, mentioned users
Task assigned	Task or backlog→task promote

Email (optional, requires SMTP in settings):

- Task / activity assignment emails
- Team schedule email (monthly from Calendar)

13. Attachments

Supported on:

- **Helpdesk issue**
- **Product backlog item**
- **Project task**

Types:

- **File upload** (stored server-side)
- **URL link**

Auto-copy on promote:

Issue —promote→ Backlog → Task → Task
attachments copy at each step

14. Linking rules (eTicket ↔ backlog)

When promoting or importing, the system avoids duplicates by matching:

Key	Helpdesk field	Backlog field
Backlog ref	<code>backlog_ref</code> (PBLID / BUGID)	<code>ref_no</code>
Client ticket	<code>external_ticket_ref</code> (No Tiket)	<code>external_ticket_ref</code>
Direct link	—	<code>issue_id</code>

Behaviour:

- If a matching backlog exists → **link** issue to it (no duplicate row).
- If not → **create** new backlog with ref from issue or auto-generated module ref (e.g. `ABB-1352`).

This supports alignment with legacy **Product Backlog Johor** / **eTicket** CSV structures without forced bulk import.

15. End-to-end example

Scenario: MBSA user cannot pay assessment tax online.

Step	Module	Action
1	Helpdesk	QA logs ticket <code>eT-CK-0123</code> , module Cukai, client MBSA, L1 assignee
2	Helpdesk	L1 checks logs → escalates to L2
3	Helpdesk	L2 confirms code defect → → Backlog on *PBT Citizen Portal*
4	Backlog	Item created Open, assignee = developer, creator = PMO who promoted
5	Backlog	Developer sets In progress , logs actual hours, comments <code>@Siti</code> for BA
6	Backlog	PMO clicks → Task "Fix payment gateway timeout"
7	Tasks	Developer completes task → Done
8	Backlog	Assignee sets Fixed → PMO sets Closed after UAT
9	Calendar	UAT session and go-live workshop logged same month
10	Delivery	UAT phase marked invoiced; Finance sees milestone

16. Module map (navigation)

Menu	Route	Role in flow
Dashboard	/	Portfolio health, open CR/bugs
Projects	/projects	Portfolio list
Project workspace	/projects/{id}	Backlog, tasks, delivery, timeline
Helpdesk	/helpdesk	Support intake
My work	/my-work	Personal queue
Calendar	/calendar	Team schedule
Gantt	/gantt	Cross-project timeline
Team	/team	People and capacity
Clients	/clients	Agency master data
Finance	/finance	Payment milestones
Reports	/reports	Exports
Users	/users	Login accounts (admin)
History	/history	Audit log
Settings	/settings/*	Locations, branding

17. Quick reference tables

Status flows side-by-side

Helpdesk	Backlog	Task
Open	Open	New
In progress	In progress	Ongoing
Waiting agency	Fixed	Done
Resolved	Closed	—

Who creates what?

Entity	Typically created by
Helpdesk ticket	PMO, support, import
Backlog item	PMO, or promote from helpdesk
Task	PMO (manual or from backlog)
Calendar activity	PMO / Admin
Project / phase	PMO / Admin

Demo data (local)

```
npm run seed:demo
npm run dev
```

Email	Password	Role
admin@pmo.local	admin123	Admin
pmo@pmo.local	pmo123	PMO
finance@pmo.local	finance123	Finance
ahmadrizal@company.com	user123	Team member

Demo seed includes sample projects, helpdesk tickets, backlog links, calendar activities, and notifications.

Related documents

Document	Description
README.md	Install, deploy, migrations
PMO-CTSB-System-Workflow.pdf	This guide as PDF
PMO-CTSB-Team-Presentation.pdf	Team slide deck
PMO-CTSB-Screenshot-Tour.pdf	UI screenshot walkthrough

Last updated: June 2026 — reflects Helpdesk → Backlog → Task workflow, backlog statuses (Open / In progress / Fixed / Closed), comments with @mentions, and calendar enhancements.